

Demo - Web Foundations

John Zimmerman

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1 Demo - Web Foundations

1.1 The Beginning

Tim Berners-Lee, a british scientist at CERN, invented to World Wide Web in 1989. Back then, the web was conceived and developed to meet the demand for automatic information-sharing between scientists in universities and institutions.

1.2 How the Internet Works

Very simply, On the client side there is a brower like Chrome of Firefox and it pushes requests to the server, and the server sends a response back to the browser. Somewhere down the line, the browser creates the **Document Object Model** (DOM) and the data (HTML/CSS) shows up visually on the screen.

1.3 The Trinity of Web Development

The three main components of web development is **HTML**, **CSS**, and **JavaScript**.

HTML HyperText Markup Language

- Builds the *structure* or skeleton of website
- Think of a frame on a house

CSS Cascading Style Sheets

- Builds the *presentation* of a website
- Think of the decorations on a house

JS JavaScript

- Builds the *behavior* and interactivity of a website
- Like the appliances of a house

1.4 The Basic Structure of a Webpage

```
<!DOCTYPE html>
<html>
<head>
  <title>Jen's Kitchen</title>
</head>

<body>
<h1>This is a heading</h1>

<h2> this is another heading</h2>

<p>This is a paragraph</p>

<br>

</body>
</html>
```

1.5 HTML Structure

Tag Refers to the descriptor or type for content. Example `<p>`, `<h1>`, etc.

Content Refers to the *stuff* in between the angle brackets of a tag.

Attribute Goes inside of a tag, used to add clarity and nuance to an element.

Value Content for the attribute.

Element The tag and the content together is the element.

If a tag doesn't have content, it's considered a **self-closing tag**. All other tags need a *closing tag*. Example. `</p>`, `</h1>`, etc.

`// examples here...`

1.6 WWW

For websites, we need a website server. This means software that handles HTTP or HTTPS. Servers job is to *serve up* documents.

1.7 IP Addresses

- Protocol :: `https://`
- Host name :: `www.`
- Domain Name :: Name of the site
- Dir path :: `/directories/`
- Doc :: absolute path - `index.html`

`https://www.nameofsite/directories/index.html`

1.8 Default files

The `index.html` is usually the file in the root of the directory and is the main document for the webpage. If you don't specify a path, it will default to `index.html`.